Customer Lifetime Value Prediction in E-Commerce: Machine Learning Approaches and Business Implications

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***Abstract*—Customer Lifetime Value (CLV) prediction has emerged as a critical component of data-driven decision-making in the e-commerce industry. This study proposes a comprehensive machine learning framework to estimate CLV using customer behavioral, transactional, and demographic data from Hepsiburada, one of Turkey's largest online marketplace. The proposed pipeline includes robust data preprocessing, multi-method feature selection, and the evaluation of over fifteen regression models via the PyCaret library. After comparative analysis, the Light Gradient Boosting Machine (LightGBM) was selected as the final model due to its strong balance of predictive accuracy and deployment efficiency. Hyperparameter tuning was conducted using Optuna, and Permutation Feature Importance was applied to reduce the feature set. The final model demonstrated high accuracy and interpretability, offering practical value for customer segmentation, targeted marketing, and strategic planning. The results highlight the potential of integrating automated machine learning pipelines with business intelligence systems to drive customer-centric growth in e-commerce.**

***Index Terms*—machine learning, e-commerce, customer lifetime value**

I. Introduction

The topic of Lifetime Value Prediction has become increasingly important for companies as a machine learning application. With these applications, companies can better understand their customers and have the opportunity to personalize the actions they take towards them. Personalized actions not only enable more effective communication with customers but also offer companies better opportunities to enhance overall customer satisfaction. In the e-commerce sector, which requires constant innovation and action, there is a significant need for such applications. Customer Lifetime Value prediction provides insights into better understanding customers, increasing long-term loyalty, and consequently driving revenue growth.

This study addresses the development and implementation of a model that predicts the lifetime value of Hepsiburada's customers for the following year. Founded in 2000, Hepsiburada is an e-commerce platform in Turkey that offers a wide range of products to its customers and prioritizes customer satisfaction by closely following technology and pioneering the best industry practices.

The model was trained using site activity data, demographic data, and purchase history of customers who have shopped at least once. Data preparation steps included filling in missing values, bucketing numerical data, label encoding, capping outliers, and feature selection to ensure data integrity.

The target variable is based on the total spending in the last 90 days, which is then annualized. The main reason for choosing this method is to ensure the recency value of customers is up-to-date and to aim for a more accurate prediction. Algorithms such as LightGBM, XGBoost, and several others were tested, and the model providing the best prediction was selected. This model contributes to obtaining customer insights that help Hepsiburada achieve its business goals and take personalized customer actions. This study serves as a concrete example of how data science and machine learning models can be used in the e-commerce sector.

A. What is CLV and Why is Important?

Customer Lifetime Value (CLV) is a predictive metric that estimates the total net profit a business can expect to generate from a customer throughout the entire duration of their relationship. Rather than focusing solely on historical transactions, CLV incorporates expected future purchases, thereby offering a forward-looking assessment of customer profitability [1]. As such, it serves as a critical analytical tool in customer relationship management, strategic marketing, and long-term revenue optimization. In environments where customer acquisition costs (CAC) are increasing and competition for consumer attention is intense—particularly in digital marketplaces—CLV becomes indispensable for ensuring the efficient allocation of marketing and operational resources.

In the context of e-commerce, the significance of CLV is further amplified. The online retail space is characterized by high customer churn rates, low switching costs, and intense price-based competition. This makes retaining existing customers and encouraging repeat purchases crucial for sustainable growth. CLV provides a quantitative foundation for segmenting customers based on long-term value rather than short-term profitability, allowing firms to prioritize high-CLV segments in their retention and personalization strategies. As highlighted by Dorrington and Goodwin, many organizations now use CLV not only to identify their most profitable customers, but also to "profile" them and "find more customers who match the profile of the most profitable customers" [1]. For instance, by identifying customers with high predicted CLVs, e-commerce platforms can allocate marketing budgets more effectively, personalize communication, and design loyalty programs that target high-value consumers. Moreover, in subscription-based business models, CLV helps forecast the average revenue expected from a customer over their subscription tenure, thereby supporting financial planning and investor reporting.

Beyond the individual customer level, CLV also holds macro-level strategic value. Investors and stakeholders often assess a firm's scalability and long-term viability through aggregated CLV metrics, using them as indicators of growth potential and customer-centric value creation. E-commerce firms can also leverage CLV insights to inform product assortment decisions—optimizing inventory and promotional strategies according to the value profiles of distinct customer cohorts. Ultimately, CLV is not just a performance indicator; it reflects a company's commitment to long-term, data-driven value generation and strategic customer management. Nevertheless, it is important to note that CLV calculations are not without challenges—modelling complexity, portfolio diversity, and unpredictable market conditions often limit accuracy. Thus, CLV should be viewed as a dynamic estimate subject to refinement, not as an exact science.

II. Literature

Customer Lifetime Value (CLV) prediction is a critical component of contemporary e-commerce strategies. It measures the total expected profitability derived from a customer over the duration of their relationship with a company. CLV predictions allow e-commerce businesses to tailor marketing strategies, optimize resource allocation, and foster long-term customer relationships, ultimately driving sustained revenue growth.

Zhuoer Ma [2] highlights the significance of leveraging data modeling techniques for accurate customer prediction, personalized marketing, and churn prediction to maximize CLV. By integrating data-driven methods such as cluster analysis, dynamic pricing, and personalized recommendations, companies can more effectively engage customers, manage assets, and optimize marketing strategies. This comprehensive approach emphasizes the proactive management of customer relationships, enabling companies to maintain competitive advantages in a data-driven environment.

Firmansyah et al. [3] underline the critical role of risk-adjusted CLV (RALTV), emphasizing that traditional CLV calculations often overlook customer risk factors like churn and income volatility. Incorporating customer risk factors enhances the precision of CLV forecasts. Their systematic literature review stresses the importance of advanced machine learning models to dynamically predict and manage customer risks, refining overall business strategies and decision-making.

Sun et al. [4] focus on machine learning algorithms and CRM analytical models, particularly within noncontractual customer relationships commonly observed in e-commerce. The authors highlight machine learning methods such as Random Forest, Support Vector Machines (SVM), and neural networks, noting their effectiveness in handling complex big data scenarios. These techniques facilitate accurate customer segmentation and robust CLV predictions, enabling businesses to target and retain high-value customers more effectively.

Norouzi [5] further expands CLV prediction methodologies by applying neural network models that integrate metrics such as Net Promoter Score (NPS), Average Transaction Value (ATV), and Customer Effort Score (CES). This research underscores the substantial predictive power of neural networks, particularly emphasizing the significant role of NPS and CES. These advanced modeling approaches enable precise customer segmentation and more personalized marketing interventions, significantly enhancing customer retention and loyalty.

Kumari et al. [6] propose utilizing deep learning models, notably Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, to capture temporal dependencies inherent in e-commerce transactional data. These advanced models surpass traditional statistical methods by effectively identifying intricate sequential patterns in customer behaviors. Leveraging such sophisticated techniques offers significant strategic advantages, supporting targeted marketing campaigns, improved retention strategies, and increased overall CLV through deeper analytical insights.

Collectively, these studies illustrate the evolution of CLV prediction methodologies, highlighting the transformative potential of machine learning and deep learning algorithms. Integrating these advanced modeling techniques into CLV analytics provides e-commerce businesses with deeper insights, enhancing strategic decision-making and enabling precise, personalized marketing strategies essential for sustained business growth.

III. Methodology

The Lifetime Value (CLV) model operates by analyzing historical customer data—such as purchase frequency, transaction amount, and recency—to estimate future customer profitability. Initially, customer data is preprocessed through feature engineering methods including handling missing values, bucketing, and encoding categorical variables. Machine learning algorithms such as LightGBM or XGBoost are then applied to predict the future monetary value of each customer. The model forecasts each customer's expected spending based on past behavior patterns, demographic attributes, and engagement metrics, allowing businesses to segment their customer base effectively, prioritize resources, and implement personalized marketing strategies to enhance customer retention and profitability.

A. Dataset

The dataset used in this study was constructed by merging tables from various sources on the Hepsiburada platform, including customer transaction histories, behavioral features, and demographic information. The primary target variable is defined as the total spending amount each customer is expected to make in the three months following the data extraction date. While calculating the target variable, extremely high spending amounts were limited with a reasonable cap to prevent outliers from adversely affecting model performance. The dataset includes all customers who made at least one purchase within the past two years. This dataset is proprietary and not publicly available. All modeling and analysis efforts were conducted on this dataset.

B. Feature Engineering

In this study, the feature engineering process was carefully designed to enhance the structural integrity of the dataset and improve the model's generalizability. Initially, columns containing missing, inconsistent, or low-information content were systematically removed. This step helped reduce dimensional complexity and mitigated the risk of overfitting.

For categorical features, standardization techniques were applied by grouping rare or non-informative categories into broader, more meaningful classes. This allowed the model to better learn class distinctions. Additionally, missing or undefined values in categorical fields were mapped to consistent placeholder categories, preventing information loss due to null values.

Numerical features were evaluated for outliers, and extreme values were either capped or transformed to minimize their disproportionate influence on the model. Variables with low variance or weak correlation with the target were also excluded to maintain a compact and informative feature set.

Collectively, these preprocessing steps streamlined the learning process by eliminating noise and redundancy. The resulting dataset included only statistically meaningful and information-rich features, leading to a model that is both accurate and interpretable.

1) Feature Selection: In this study, a comprehensive feature selection process was employed to identify the most relevant predictors for the CLV model. Four different techniques were utilized: Boruta, Recursive Feature Elimination (RFE), SelectKBest with f-regression, and LightGBM-based feature importance. Each method was applied independently, and their results were consolidated into a scoring table, where features could receive a maximum score of 4 and a minimum of 0, based on their selection by each method. Only features with a score of at least 3 were included in the final model to ensure both predictive strength and model interpretability.

- **Boruta** algorithm, which builds upon the Random Forest technique, identifies all relevant features by comparing the importance of actual predictors to their randomized counterparts (shadow features). Through iterative elimination of features deemed less important than noise, Boruta effectively isolates informative variables, improves model interpretability, and helps prevent overfitting—especially in high-dimensional datasets [7].
- **Recursive Feature Elimination (RFE)** sequentially removes the least important features from the dataset, based on the model's performance at each iteration. Beginning with the complete feature set, RFE fits the model, ranks features according to their importance, and recursively eliminates the lowest-ranked ones. This method ultimately selects an optimal subset of features, supporting model simplicity and predictive accuracy, and is particularly useful for filtering out irrelevant features in complex datasets [8].
- **SelectKBest** combined with the f-regression test statistically assesses the relationship between each independent variable and the target variable. The F-statistic quantifies the explanatory power of each feature, with the top-scoring predictors being retained. This method enhances both model interpretability and predictive performance, especially in multivariate regression contexts.
- **LightGBM feature importance** assesses each variable's contribution to the predictive model, ranking features according to their calculated importance. Features that surpass a defined importance threshold—often based on the distribution of feature importances—are selected. This process reduces model complexity, limits the influence of noisy variables, and improves the robustness and generalizability of the final model, particularly for large and complex datasets. [9]

As a result of this multi-method feature selection strategy, the initial set of numerous features was reduced to a refined subset of 48 features. This dimensionality reduction not only improved computational efficiency but also enhanced the model's robustness, interpretability, and predictive performance.

IV. Machine Learning Model

A. Model Selection with PyCaret

To efficiently explore a wide range of regression models, the PyCaret library was employed. PyCaret is an open-source, low-code machine learning framework that enables end-to-end model development with minimal coding effort [10]. It provides a standardized interface to test and compare models under consistent preprocessing and validation pipelines. In this study, more than fifteen regression models were evaluated using PyCaret, including linear models (e.g., Linear Regression, Ridge Regression, Lasso), regularization-based techniques (e.g., Elastic Net, Least Angle Regression), ensemble learners (e.g., Random Forest, Gradient Boosting, AdaBoost, Extra Trees), and other algorithms such as Bayesian Ridge, K Nearest Neighbors, and LightGBM. Each model was trained and validated using cross-validation, and their performance metrics were compared to identify the most suitable candidate.

B. LightGBM and Hyperparameter Optimization with Optuna

Among the tested models, Light Gradient Boosting Machine (LightGBM) demonstrated superior performance in terms of predictive accuracy and computational efficiency. LightGBM is a gradient boosting framework that constructs decision trees in a leaf-wise manner, allowing for faster convergence and reduced loss compared to traditional level-wise growth [11]. Its ability to handle large-scale datasets and support for categorical features without explicit one-hot encoding makes it well-suited for high-dimensional structured data. To further enhance model performance, hyperparameter tuning was carried out using the Optuna framework [12]. Optuna is a next-generation hyperparameter optimization library that employs efficient sampling algorithms and pruning techniques based on Bayesian optimization. Key hyperparameters such as learning rate, number of leaves, max depth, subsample ratio, feature fraction, and regularization terms were optimized through multiple trials. This tuning process significantly improved the model's generalization ability and reduced overfitting risk.

C. Feature Selection with Permutation Importance

Following the hyperparameter optimization of the LightGBM model, a feature selection process was implemented to reduce dimensionality and improve interpretability. The Permutation Feature Importance method was used, which measures the change in model performance when the values of each feature are randomly permuted [13]. This approach effectively identifies features that have the most substantial influence on the target variable by quantifying the performance degradation caused by disrupting each feature's signal.

Using this method, the top six most important features were selected from the set of 48 feature which selected in the first phase. This dimensionality reduction preserved the predictive power of the model while simplifying its structure. The final LightGBM model was then retrained and re-optimized using only these six features, resulting in a leaner model that maintained high accuracy, improved inference speed, and enhanced interpretability.

V. Results

A. Model Evaluation Metrics

Figure-1 presents the performance of various regression models tested during the CLV prediction task. The metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Training Time in seconds (TT).

B. Model Comparison and LightGBM Selection

As shown in Figure-1, the Gradient Boosting Regressor (GBR) achieved the best performance in terms of MSE and RMSE, with values of 3,568,853.49 and 1888.48 respectively. However, the Light Gradient Boosting Machine (LightGBM) achieved the lowest MAE (286.58) and demonstrated nearly identical RMSE (1889.77), with only a marginal trade-off in MSE. Despite GBR's slight edge in RMSE, LightGBM was selected as the final model based on a broader set of considerations. LightGBM offers several practical advantages in production environments, including native handling of categorical variables, leaf-wise tree growth that enhances convergence, and efficient memory usage [11]. Additionally, its widespread adoption and strong community support make it a sustainable choice for both experimentation and deployment. While GBR had faster training time (20.45 seconds) compared to LightGBM (176.58 seconds), the additional training overhead of LightGBM was acceptable given the improved maintainability and scalability in a real-world system. Other models, such as AdaBoost, Ridge, and Lasso regressors, yielded significantly higher RMSE values above 1900 and did not offer any meaningful performance trade-offs.

Model	MAE	MSE	RMSE	TT (Sec)
Gradient Boosting Regressor	287.14	**3,568,853.49**	**1,888.48**	20.46
Light Gradient Boosting Machine	**286.58**	3,573,759.14	1,889.77	176.58
AdaBoost Regressor	294.79	3,612,381.13	1,899.98	4.44
Linear Regression	294.58	3,644,038.31	1,908.25	3.83
Lasso Regression	294.93	3,644,281.30	1,908.31	0.48
Ridge Regression	294.58	3,644,038.34	1,908.25	0.21
Least Angle Regression	294.58	3,644,038.31	1,908.25	0.21
Lasso Least Angle Regression	294.93	3,644,281.30	1,908.31	0.21
Bayesian Ridge	294.58	3,644,038.49	1,908.25	0.24
Elastic Net	353.84	3,686,282.32	1,919.30	0.32
Orthogonal Matching Pursuit	342.56	3,852,575.12	1,962.13	**0.20**
Random Forest Regressor	305.83	3,908,561.07	1,976.44	28.53
K Neighbors Regressor	287.77	4,213,630.90	2,051.95	1.84
Extra Trees Regressor	310.72	4,400,969.52	2,097.28	22.52
Decision Tree Regressor	319.91	6,348,963.91	2,519.38	1.21
Passive Aggressive Regressor	493.55	16,927,444.21	3,432.92	0.45

Fig. 1. Results

Notably, models like Elastic Net and Orthogonal Matching Pursuit showed weaker results in both error and training efficiency, while tree-based methods like Random Forest and Extra Trees underperformed in accuracy and were computationally more expensive. The Passive Aggressive Regressor demonstrated extremely poor performance across all metrics. In conclusion, LightGBM was selected as the optimal model for its strong balance between predictive performance, robustness, and production viability.

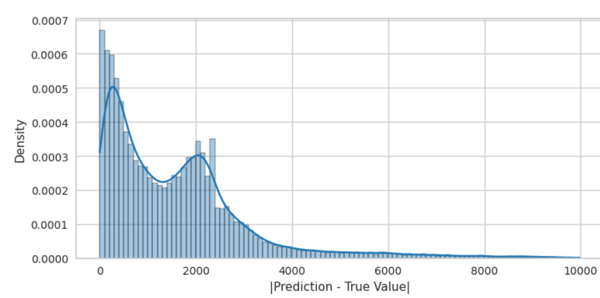

Fig. 2. LightGBM prediction absolute errors, greater than 0 to 10,000

Figure-2 presents the distribution of absolute errors between LightGBM predictions and true values, specifically within the range of greater than 0 to 10,000. The concentration of errors near the lower end highlights the model's strong predictive accuracy for most cases. However, the presence of a long tail indicates occasional larger discrepancies, suggesting areas for potential model improvement. This pattern underscores the model's effectiveness while also pointing to opportunities for refining predictions in outlier scenarios.

VI. Conclusion & Future Research

In this study, we presented a comprehensive machine learning pipeline for predicting Customer Lifetime Value (CLV) in the context of e-commerce, using data obtained from Hepsiburada, one of Turkey's leading online retail platform. Our approach involved a multi-stage modeling process that included extensive feature engineering, multi-method feature selection, and the evaluation of more than fifteen regression models via PyCaret. Among these, LightGBM emerged as the most effective model due to its strong predictive performance, computational efficiency, and robustness in production environments.

To further optimize the LightGBM model, hyperparameter tuning was conducted using the Optuna framework. Additionally, the application of Permutation Feature Importance enabled us to reduce the feature space from 48 to 6 highly informative predictors without sacrificing accuracy. This reduction significantly enhanced the model's interpretability, inference speed, and overall maintainability.

The results of our model comparisons demonstrated that LightGBM achieved the lowest MAE while maintaining comparable RMSE to other top-performing models. Despite having a longer training time than some alternatives, LightGBM's overall advantages in deployment-readiness and support for categorical features made it the most viable choice for real-world use.

Looking ahead, several directions remain for future research. First, the integration of sequential deep learning architectures, such as Recurrent Neural Networks (RNNs) or Transformer-based models, may further enhance the model's ability to capture temporal patterns in customer behavior. Second, introducing macroeconomic indicators or external market variables could offer improved predictive insights. Lastly, deploying the model in a live environment and integrating it with automated marketing workflows would allow for real-time customer segmentation and personalization, potentially boosting both retention and revenue.

In conclusion, this work demonstrates the tangible benefits of combining domain-specific data with state-of-the-art machine learning techniques to predict CLV in e-commerce. The proposed pipeline offers a scalable and interpretable solution that can support strategic decision-making and long-term customer relationship management.

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